Zomato Data Analysis

1. Project Overview

The Zomato Data Analysis focuses on analyzing restaurant data to understand customer ordering behaviour, restaurant popularity, and rating patterns on the Zomato platform. With the rapid growth of online food delivery services, platforms like Zomato collect large amounts of data related to restaurants and customer interactions. Analysing this data helps identify trends such as the most popular restaurant types, customer spending habits, and the relationship between online ordering and customer ratings.

2. Dataset Summary

The dataset consists of multiple rows and columns where:

Rows represent individual restaurants.
Columns represent different attributes related to restaurant services and customer feedback.

3. Exploratory Data Analysis

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
df = pd.read_csv("Zomato data .csv")
print(df)
```

```
                     name online_order book_table   rate  votes  \
0                   Jalsa          Yes        Yes  4.1/5    775
1           Spice Elephant          Yes         No  4.1/5    787
2          San Churro Cafe          Yes         No  3.8/5    918
3    Addhuri Udupi Bhojana           No         No  3.7/5     88
4            Grand Village           No         No  3.8/5    166
..                     ...          ...        ...    ...    ...
143       Melting Melodies           No         No  3.3/5      0
144        New Indraprasta           No         No  3.3/5      0
145           Anna Kuteera          Yes         No  4.0/5    771
146                 Darbar           No         No  3.0/5     98
147          Vijayalakshmi          Yes         No  3.9/5     47

     approx_cost(for two people) listed_in(type)
0                            800          Buffet
1                            800          Buffet
2                            800          Buffet
3                            300          Buffet
4                            600          Buffet
..                           ...             ...
143                          100          Dining
144                          150          Dining
145                          450          Dining
146                          800          Dining
147                          200          Dining

[148 rows x 7 columns]
```

```
df.describe()
```

	votes	approx_cost(for two people)
count	148.000000	148.000000
mean	264.810811	418.243243
std	653.676951	223.085098
min	0.000000	100.000000
25%	6.750000	200.000000
50%	43.500000	400.000000
75%	221.750000	600.000000
max	4884.000000	950.000000

- **Convert data type:** The convert data type of the 'rate' column to float and remove denominator.

```
def handlerate(value):
    value = str(value).split('/')
    value = value[0]
    return float(value)
df['rate'] =df['rate'].apply(handlerate)
print(df.head())
```

```
                    name online_order book_table  rate  votes  \
0                  Jalsa          Yes        Yes   4.1    775
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   approx_cost(for two people) listed_in(type)
0                          800          Buffet
1                          800          Buffet
2                          800          Buffet
3                          300          Buffet
4                          600          Buffet
```
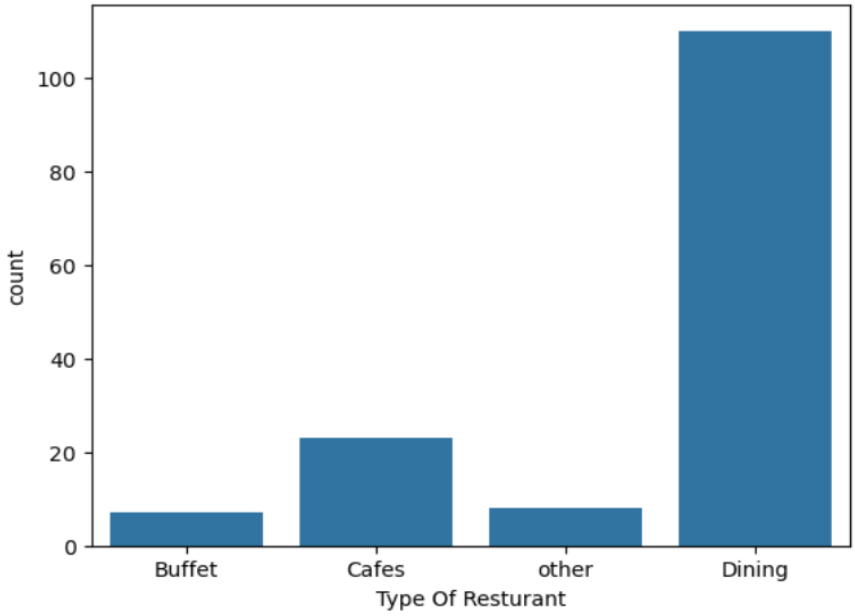

4. Data Visualization

- **Types of Restaurants:**
  Check to what types of restaurants do the majority of customer order from?

```
Types of Resturant
```

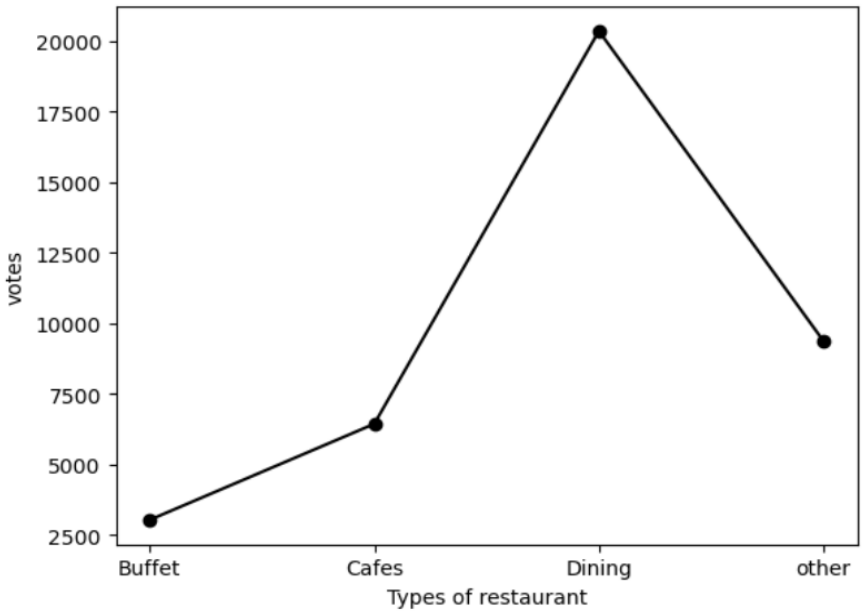
```
sns.countplot(x =df['listed_in(type)'])
plt.xlabel('Type Of Resturant')
plt.show()
```



- **Voting by each type of restaurant:**
  How many votes has each type of restaurant received from customer?

```
grouped_data = df.groupby('listed_in(type)')['votes'].sum()
result = pd.DataFrame({'votes':grouped_data})
plt.plot(result,c ='Black',marker ='o')
plt.xlabel("Types of restaurant",c = 'black',size = 10)
plt.ylabel('votes',c = 'black',size = 10)
```

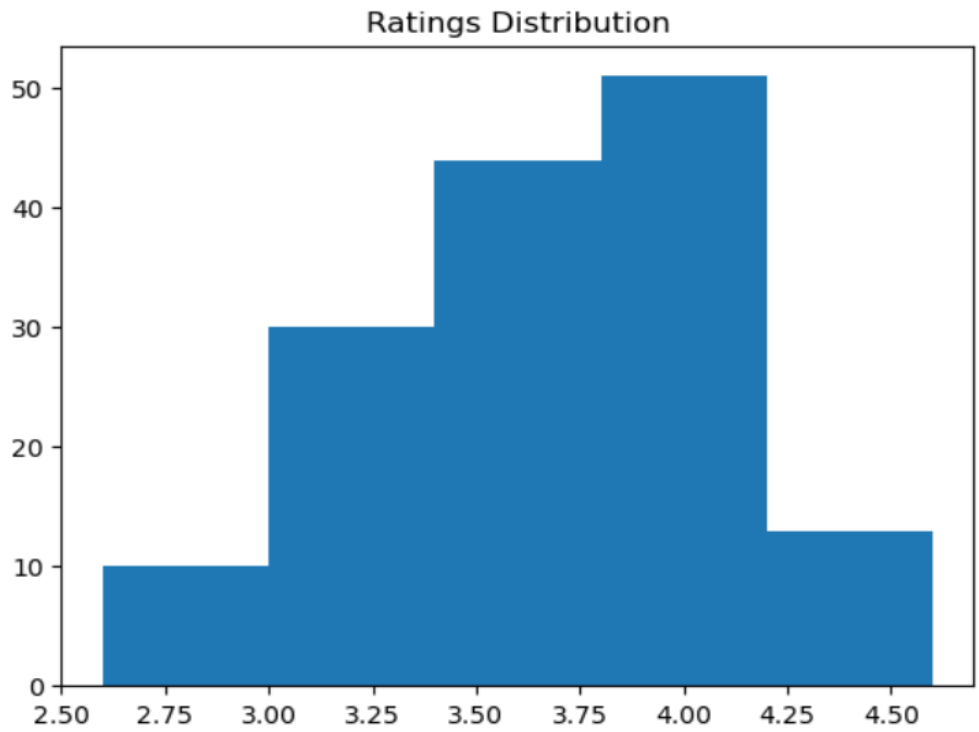
```
Text(0, 0.5, 'votes')
```



- **Ratings:**

  What are the ratings that the majority of restaurants received from customers?

```
plt.hist(df['rate'],bins = 5)
plt.title("Ratings Distribution")
plt.show()
```

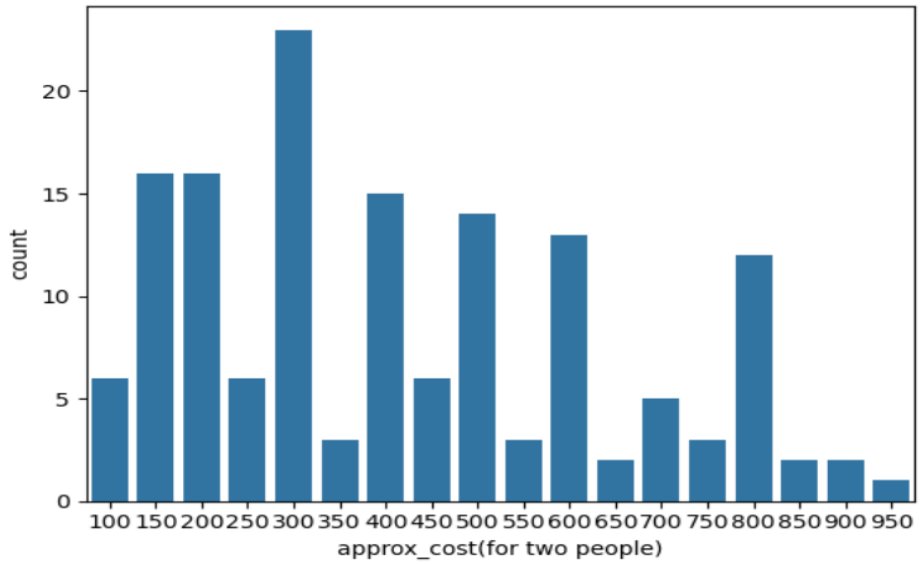


- **Order By Couple:**

  Observed that most couple order most their food online and their average spending on each order.

```
couple_data = df['approx_cost(for two people)']
sns.countplot(x = couple_data)
```

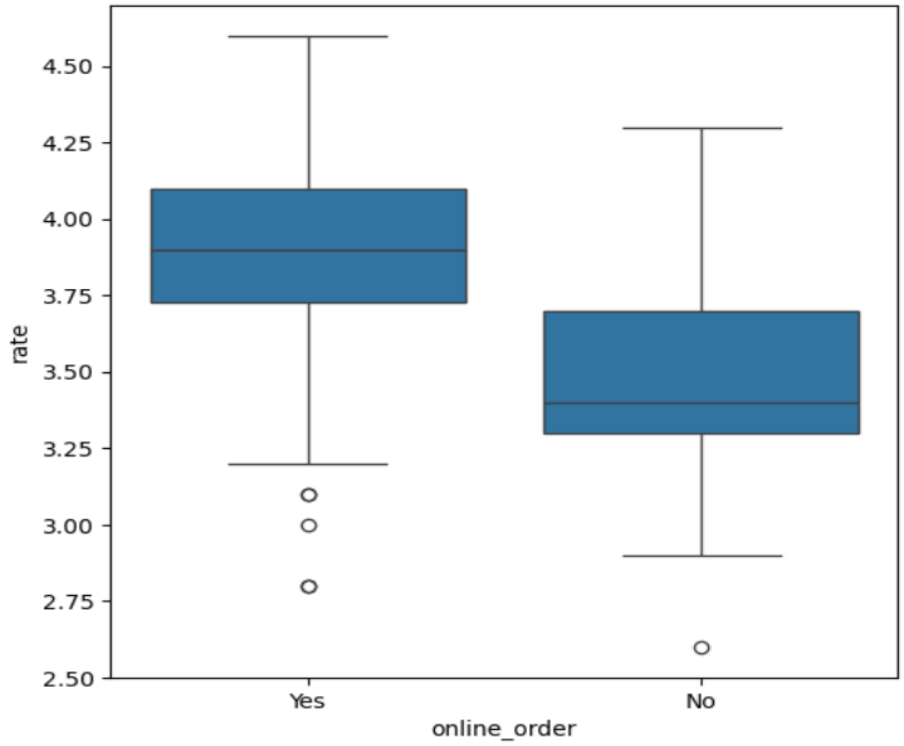
```
<Axes: xlabel='approx_cost(for two people)', ylabel='count'>
```

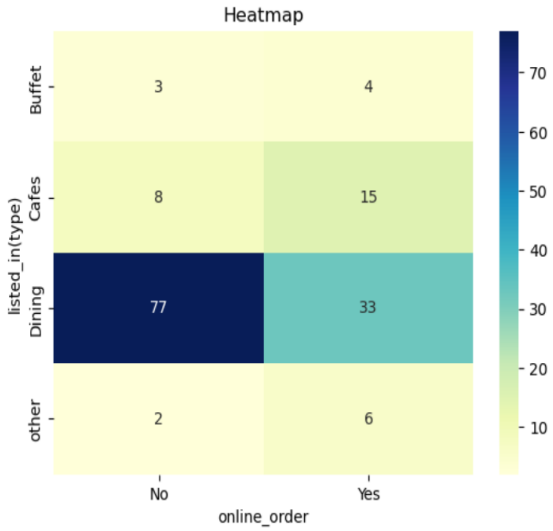


- **Online VS Offline Ratings:**
  Which mode has received the maximum ratings?

```
plt.figure(figsize =(6,6))
sns.boxplot(x = 'online_order', y = 'rate',data = df)
```

```
<Axes: xlabel='online_order', ylabel='rate'>
```



- **Restaurant received more offline orders:**
  Which types of restaurants received more offline orders, so that Zomato can provide those customers with some good offers?

```
pivot_table=df.pivot_table(index='listed_in(type)',columns = 'online_order',aggfunc = 'size',fill_value = 0)
sns.heatmap(pivot_table,annot = True,cmap= 'YlGnBu',fmt ='d')
plt.title("Heatmap")
plt.xlabel = ("Online Order")
plt.ylabel =("Listed In (Type)")
plt.show()
```